

1 State Representation and Bra-ket algebra

In Quantum Computing, you'd be dealing a lot with matrices, as most operations are essentially linear algebra operations. The term 'vector' by default would be referring to the column vector, which can be represented in the following formats: $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ or $(1,2)$.

A row vector would be represented as $[1 \ 2]$ or as a shorthand notation- $(1,2)^T$ or $(1,2)'$. These are some of the standard quantum mechanical notations for notions in linear algebra. Don't fret over any new terms encountered, I'll be explaining the terms mentioned and related concepts.

Notation	Description
z^*	Complex conjugate of the complex number z . $(1+i)^* = 1-i$
$ \psi\rangle$	Known as ket. it is a column vector
$\langle\psi $	Known as bra. It is a row vector and a vector dual to $ \psi\rangle$
$\langle\phi \psi\rangle$	Inner product between the vectors $ \phi\rangle$ and $ \psi\rangle$.
$ \phi\rangle \otimes \psi\rangle$	Tensor product of $ \phi\rangle$ and $ \psi\rangle$.
$ \phi\rangle \psi\rangle$ or $ \phi\psi\rangle$	Abbreviated notation for tensor product of $ \phi\rangle$ and $ \psi\rangle$.
A^*	Complex conjugate of the A matrix.
A^T	Transpose of the A matrix.
$A^\dagger$	(Read A dagger) Hermitian conjugate or adjoint of the A matrix, $A^\dagger = (A^T)^*$.
	$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}$
$\langle\phi A \psi\rangle$	Inner product between $ \phi\rangle$ and $A \psi\rangle$. Equivalently, inner product between $A^\dagger \phi\rangle$ and $ \psi\rangle$.

1.1 State Representation

The two computational basis states for a qubit are $|0\rangle$ and $|1\rangle$, which can be represented using matrix as

$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. A qubit can be in a state other than $|0\rangle$ and $|1\rangle$, which is what makes it fundamentally different from classical bits. It can exist in a superposition of these basis states, a qubit can be a linear combination of the computational basis states :

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are complex numbers and When we measure a qubit we get either the result 0, with probability $|\alpha|^2$ or the result 1, with probability $|\beta|^2$.

Naturally,

$$|\alpha|^2 + |\beta|^2 = 1.$$

since the probabilities must sum to one.

For a 2-qubit system, the 4 computational basis states are $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. Similar to the case of single qubits, a pair of qubits can exist as a superposition of these basis states:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

Similarly, for an n-qubit system, there would be 2^n computational basis states. For a multi-qubit system, the qubits are combined using tensor product.

1.2 Bra-Ket algebra

As mentioned in the above table, $|\psi\rangle$ is the ket notation and $\langle\psi|$ is the bra notation. Some other properties related to Bra-Ket algebra are:

- The ket ($|\psi\rangle$) and its corresponding bra ($\langle\psi|$) are complex conjugate transposes of each other . i.e $|\psi\rangle^\dagger = \langle\psi|$.
- The inner product of a state with itself gives its squared magnitude, which must equal 1 for valid probability distributions: $\langle\psi|\psi\rangle = 1$
- Two states are said to be orthogonal if their inner product is 0 : $\langle\psi|\phi\rangle = 0$

You can read in detail about inner products [here](#)

1.3 Tensor Product

Tensor product is used to define and combine multiple qubits.

- For 2 matrices A and B : $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 5 \\ 1 & 2 \end{bmatrix}$

You multiply each element of Matrix A by the entire Matrix B to obtain the tensor product:

$$A \otimes B = \left[\begin{array}{cc} 1 \times \begin{bmatrix} 0 & 5 \\ 1 & 2 \end{bmatrix} & 2 \times \begin{bmatrix} 0 & 5 \\ 1 & 2 \end{bmatrix} \\ 3 \times \begin{bmatrix} 0 & 5 \\ 1 & 2 \end{bmatrix} & 4 \times \begin{bmatrix} 0 & 5 \\ 1 & 2 \end{bmatrix} \end{array} \right] \text{ This results in a larger } 4 \times 4 \text{ matrix: } A \otimes B = \begin{bmatrix} 0 & 5 & 0 & 10 \\ 1 & 2 & 2 & 4 \\ 0 & 15 & 0 & 20 \\ 3 & 6 & 6 & 12 \end{bmatrix}$$

- The state $|00\rangle$ which is a tensor product of $|0\rangle$ and $|0\rangle$ can be represented the following way:

$$|0\rangle \otimes |0\rangle = \left[\begin{array}{c} 1 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ 0 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{array} \right] \text{ which is equal to the matrix } \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$